the neighboring, ‘negative-energy regions of space. The photons, now real, are liberated from each other. One is inside the horizon and lost forever from the external Universe. The other escapes from the hole, carrying away the energy (that is, the mass[13]) that the hole’s tidal gravity gave to it. The hole, with its mass reduced, shrinks a bit.

This mechanism of emitting particles does not depend at all on the fact that the particles were photons, and their associated waves were electromagnetic. The mechanism will work equally well for all other forms of particle/wave (that is, for all other types of radiation—gravitational, neutrino, and so forth), and therefore a black hole radiates *all* types of radiation.

Before the virtual particles have materialized into real particles, they must stay closer together than roughly the wavelength of their waves. To acquire enough energy from the hole’s tidal gravity to materialize, however, they must get as far apart as about a quarter of the circumference of the hole. This means that the wavelengths of the particle/waves that the hole emits will be about one-fourth the hole’s circumference in size, and larger.

A black hole with mass twice as large as the Sun has a circumference of about 35 kilometers, and thus the particle/waves that it emits have wavelengths of about 9 kilometers and larger. These are enormous wavelengths compared to light or ordinary radio waves, but not much different from the lengths of the gravitational waves that the hole would emit if it were to collide with another hole.

During the early years of his career, Hawking tried to be very careful and rigorous in his research. He never asserted things to be true unless he could give a nearly airtight proof of them. However, by 1974 he had changed his attitude: “I would rather be right than rigorous,” he told me firmly. Achieving high rigor requires much time. By 1974 Hawking had set for himself goals of understanding the full marriage of general relativity with quantum mechanics, and understanding the origin of the Universe—goals that to achieve would require enormous amounts of time and concentration. Perhaps feeling more finite than other people feel because of his life-shortening disease, Hawking felt he could not afford to dally with his discoveries long enough to achieve high rigor, nor could he afford to